IILM UNIVERSITY, GREATER NOIDA

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

TECHNICAL DOCUMENTATION

Project Title: Predictive Maintenance System for Data Centre Servers

1. Project Details

Project ID	26P_3CSE276-Predictive Maintenance System for Data Centre Servers
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2. Introduction

The Predictive Maintenance System for Data Centre Servers uses server monitoring data (CPU usage, memory usage, temperature and disk usage) to identify potential server issues and give a predictive maintenance indication or alert.

Note: This describes the implementation in pm.py and run.py. The training data is synthetic; see the Dataset and Results Report.

3. System Objectives

- Accept server monitoring records as input.
- Validate each record for missing, non-numeric and out-of-range values.
- Predict whether a server needs maintenance with a trained classifier.
- Return a status (NORMAL or MAINTENANCE ALERT) with a probability.

4. System Architecture

Pipeline: Data Input → Validation → Prediction Model → Result (status and probability).

Component	Responsibility
Data generation	generate_dataset() creates synthetic server telemetry and labels.
Validation	validate() checks each field is present, numeric and within its valid range; raises ValidationError otherwise.
Prediction	predict() validates one record and returns prediction, status and probability from the trained model.
Batch processing	predict_batch() processes many records and reports invalid ones as errors without stopping.

Component	Responsibility
Training and evaluation	run.py trains three models, evaluates them, selects the best by F1 and runs the test cases.

5. Technology Stack

Language	Python 3.13.18
Libraries	scikit-learn 1.9.1, pandas 3.0.5, NumPy 2.5.3, joblib
Model	Gradient Boosting classifier (saved as model.joblib)
Interface	Python functions and command-line script; no graphical interface
Environment	Linux 6.18.44-fc-v185 (cloud VM)

6. Data and Model

- Dataset: server_telemetry_synthetic.csv, 5,000 records (synthetic).
- Input features: cpu_usage, memory_usage, temperature, disk_usage.
- Target: maintenance_needed (1 = needs maintenance).
- Model: Gradient Boosting; 80/20 stratified split; threshold 0.5.
- Result on test set: accuracy 87.70%, F1 0.817.

7. Setup and Usage

- Install dependencies: pip install scikit-learn pandas numpy joblib
- Train, evaluate and test: python run.py (writes the dataset CSV, model.joblib and results.json).
- Predict for one record: load model.joblib with joblib, then call pm.predict(model, {"cpu_usage": 90, "memory_usage": 85, "temperature": 80, "disk_usage": 70}).
- Batch prediction: pm.predict_batch(model, list_of_records).

8. Validation Rules

Field	Valid Range	Behaviour on invalid input
cpu_usage	0 – 100	Missing, non-numeric or out-of-range value raises ValidationError
memory_usage	0 – 100	Same
temperature	10 – 110 °C	Same
disk_usage	0 – 100	Same

9. Limitations and Future Work

- Trained on synthetic data; performance on real servers is unknown.

- Only four metrics are used; real systems also expose fan speed, power draw, error logs and SMART disk data.
- Possible future work: real-time monitoring, retraining on real telemetry, threshold tuning to reduce missed failures, and a dashboard or alert integration.

10. Related Documents

- Project Testing Report
- Performance Results Report
- Dataset and Results Report